

Yutong (Richie) Jin

Auckland, New Zealand | +64 027 292 2743 | richiejyt@gmail.com

Professional Summary

Master of Engineering Graduate and HCI & VR Researcher combining a strong Computer Science background with expertise in **VR Development**, **Game Design**, and **Machine Learning**. Experienced in creating immersive educational environments with integrated multi-modal AI agents, alongside a solid history of deploying ML models for healthcare and energy sectors. Proficient in Unity, Python, and Full-stack development, committed to delivering innovative and optimized technical solutions.

Education

University of Canterbury (GPA: 7.63) <i>Master of Engineering in Human Interface Technology</i>	Christchurch, New Zealand Feb 2025 – Apr 2026
Yanbian University <i>Bachelor of Science in Computer Science</i>	Yanji, Jilin, China Jun 2017 – Sep 2021

Professional Experience

HITLab NZ, University of Canterbury <i>Research Assistant</i>	Christchurch, New Zealand Nov 2025 – Jan 2026
- Integrated advanced multi-modal AI models (LLM, VLM, TTS, STT) into a Unity-based educational project to enhance interactive learning experiences. - Designed and implemented a robust full-stack architecture featuring a Python server backend and a responsive Unity frontend. - Created comprehensive tutorials and documentation to facilitate beginners' understanding of C# programming and project codebase.	
WEL Networks <i>Software Engineer Intern</i>	Christchurch, New Zealand Jun 2025 – Jul 2025
- Developed a Python-based inverter compliance detection system, processing over 6 million smart meter records to enhance grid data analysis. - Implemented algorithms for Export Limit, Volt-Watt, and Volt-Var compliance detection. - Delivered an automated tool that replaced manual verification processes, enabling real-time monitoring of grid stability.	
Shenzhen Institute of Advanced Technology (SIAT), CAS <i>Machine Learning Engineer</i>	Shenzhen, China Sep 2022 – Oct 2023
- Engineered a Machine Learning system for core body temperature prediction, achieving 95% accuracy and deployed for real-time monitoring. - Built a production-ready exercise fatigue detection platform using deep learning on ECG/IMU data (85.46% accuracy). - Optimized models (XGBoost, SVR, Neural Networks) for edge deployment, successfully reducing inference time by 40%.	

Academic Projects

Master's Thesis Project: Intelligent VR Game Development with Multi-Modal AI Agents
- Spearheaded the redesign and development of three immersive VR games, each featuring a dedicated autonomous AI agent to assist users. - Seamlessly integrated cutting-edge multi-modal AI technologies (LLM, STT, TTS) to enable natural, real-time voice and visual interactions between players and agents. - Executed a rigorous between-subjects user study with 26 participants (13 per condition) to evaluate the longitudinal impact of identity continuity on human-agent relationship formation. - Formulated statistical analysis workflows using Python, utilizing Analysis of Covariance (ANCOVA) to control baseline group imbalances and validate significant improvements in rapport, emotional connectedness, and credibility.
COVID-19 Vaccine Response Analysis Shenzhen, China
- Analyzed a dataset of 50,000+ vaccination records using ensemble ML methods (Random Forest, Neural Networks) to identify trends. - Collaborated with Dr. Yunpeng Cai to identify statistical correlations between vaccines and reported symptoms.

Skills & Awards

Programming & Frameworks: C#, Python, Java, R, Swift, PHP, MATLAB, SQL.

Tools & Platforms: Unity, Kubernetes, Docker, VMware, Linux, HTML.

Languages: English (Fluent), Korean (Fluent), Chinese (Native)